ALGEBRA 2 · UNIT 9: CONIC SECTIONS

Signals & Dishes

Use circles, parabolas, ellipses, and hyperbolas to design communication technology.

Honors

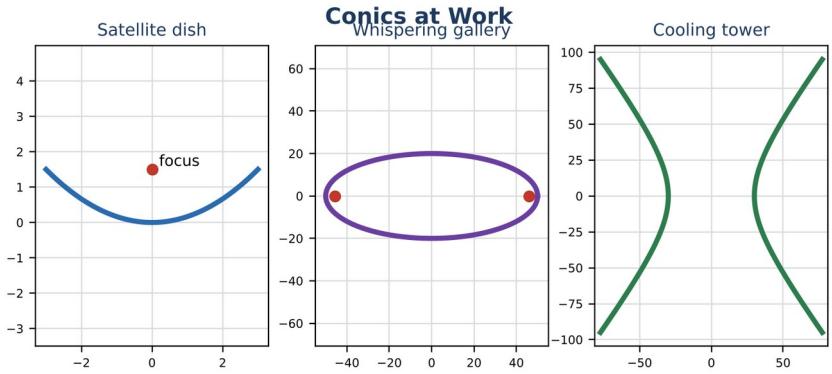
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

SignalWorks builds cell towers, satellite dishes, and even whispering galleries for science museums. Each design depends on one of the conic sections.

You will analyze each design with its equation, then create a coverage plan for a new town.



Learning Goals

- I can derive conic equations from geometric definitions.
- I can use distance differences to locate points (hyperbolas).
- I can optimize a design.

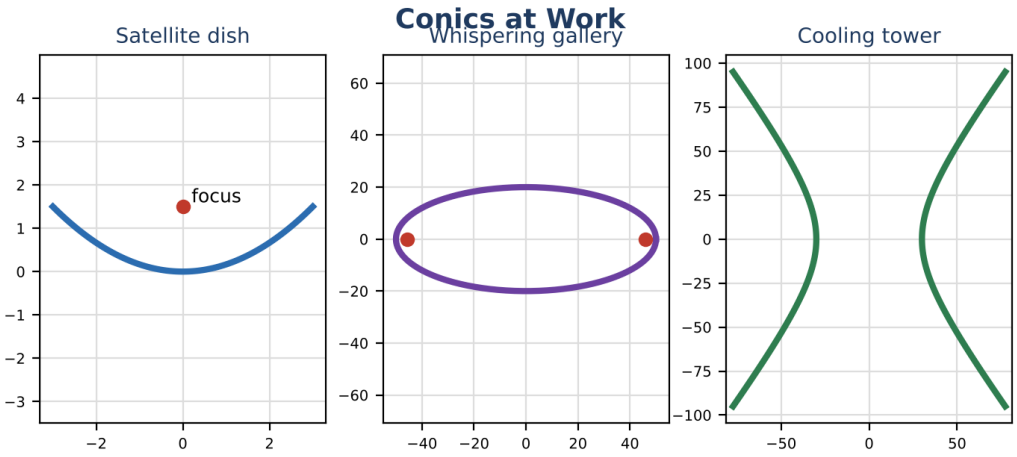
Project Timeline

When	What you will do
Day 1	Definitions
Day 2	Locating a Ship
Day 3	Optimized Coverage
Final	Engineering Design Brief + reflection

Part 1: Definitions

Day 1 · Use the distance formula.

1. Derive the equation of the parabola with focus (0, p) and directrix y = −p.



2. A dish is 10 ft across and 1.25 ft deep. Find the focus.

3. Show that every point on $x^2 \div 2500 + y^2 \div 400 = 1$ has distances to the foci that add to 100.

Part 2: Locating a Ship

Day 2 · Two stations are at (−50, 0) and (50, 0) (miles). A ship's signal arrives at the right station first, and the difference in distances is 60 miles.

4. Explain why the ship is on a hyperbola and write its equation.

5. The ship is also on the line y = 32. Find its location (on the right branch).

6. Write the asymptotes and explain what they mean far from the stations.

Part 3: Optimized Coverage

Day 3 · Houses: (0, 0), (4, 2), (−3, 5), (6, −4), (8, 3).

7. Find the smallest single circle that covers all houses, or show two towers are needed with radius 5.

8. Write the equations and test every house.

9. Write a brief explaining the design.

Final Product: Engineering Design Brief

Your final product must include:

- Derivations
- Ship location
- Optimized coverage plan
- Brief

Reflection

R1. Why is a hyperbola the set of points with a constant difference of distances?

R2. How do conic definitions lead to real designs?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Coverage plan uses correct circle equations and point tests, and every conic feature is correct. Goes beyond the minimum with a creative, realistic choice.	Coverage plan uses correct circle equations and point tests, and every conic feature is correct.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Brief includes each conic's equation, key features (center, foci, vertices, radius), and a coverage map. Polished, organized, and visually engaging.	Brief includes each conic's equation, key features (center, foci, vertices, radius), and a coverage map. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.